Loan Default & Credit Risk Analysis

Loan Default & Credit Risk Analytics Dashboard

Project Report

End-to-end analytics project using Python, Pandas, Jupyter/VS Code, Power BI and DAX.

Cleaned dataset: 8,000 records

1. Executive Summary

The cleaned dataset contains 8,000 loans from 8,000 customers. There are 1,126 defaulted loans and 6,874 non-defaulted loans, producing an overall default rate of 14.07%. Average annual income is $54,913.69, average credit score is 660.13, and average loan amount is $80,127.50.

Metric	Value	Metric	Value
Total Customers	8,000	Total Loans	8,000
Defaulted Loans	1,126	Non-Defaulted Loans	6,874
Default Rate	14.07%	Average Loan Amount	$80,127.50
Average Credit Score	660.13	Average Annual Income	$54,913.69
Average Interest Rate	10.44%	Average DTI	37.58

2. Objectives & Methodology

- Analyze overall loan-default distribution and default rate.
- Compare default rates by loan type and employment status.
- Examine default-rate patterns across credit-score and income groups.
- Analyze average loan amounts by loan type and region.
- Explore annual income versus loan amount.
- Provide an interactive loan-level risk explorer.

Data preparation included numeric type conversion, median handling of missing numeric values in the cleaning workflow, and removal of duplicate customer records while retaining the first occurrence. The final cleaned dataset contains 8,000 records.

Credit-score groups are Poor (<580), Fair (580-669), Good (670-739), and Excellent (740+). Dashboard income groups are Low (<30,000), Medium (30,000-59,999), High (60,000-99,999), and Very High (100,000+).

3. Descriptive Findings

Default Rate by Loan Type

Loan Type	Default Rate
Business Loan	32.50%
Personal Loan	12.57%
Home Loan	10.41%
Education Loan	9.32%
Auto Loan	7.33%

Credit Score Group Default Rates

Credit Score Group	Default Rate
Poor (<580)	34.26%
Fair (580-669)	16.16%
Good (670-739)	7.00%
Excellent (740+)	3.12%

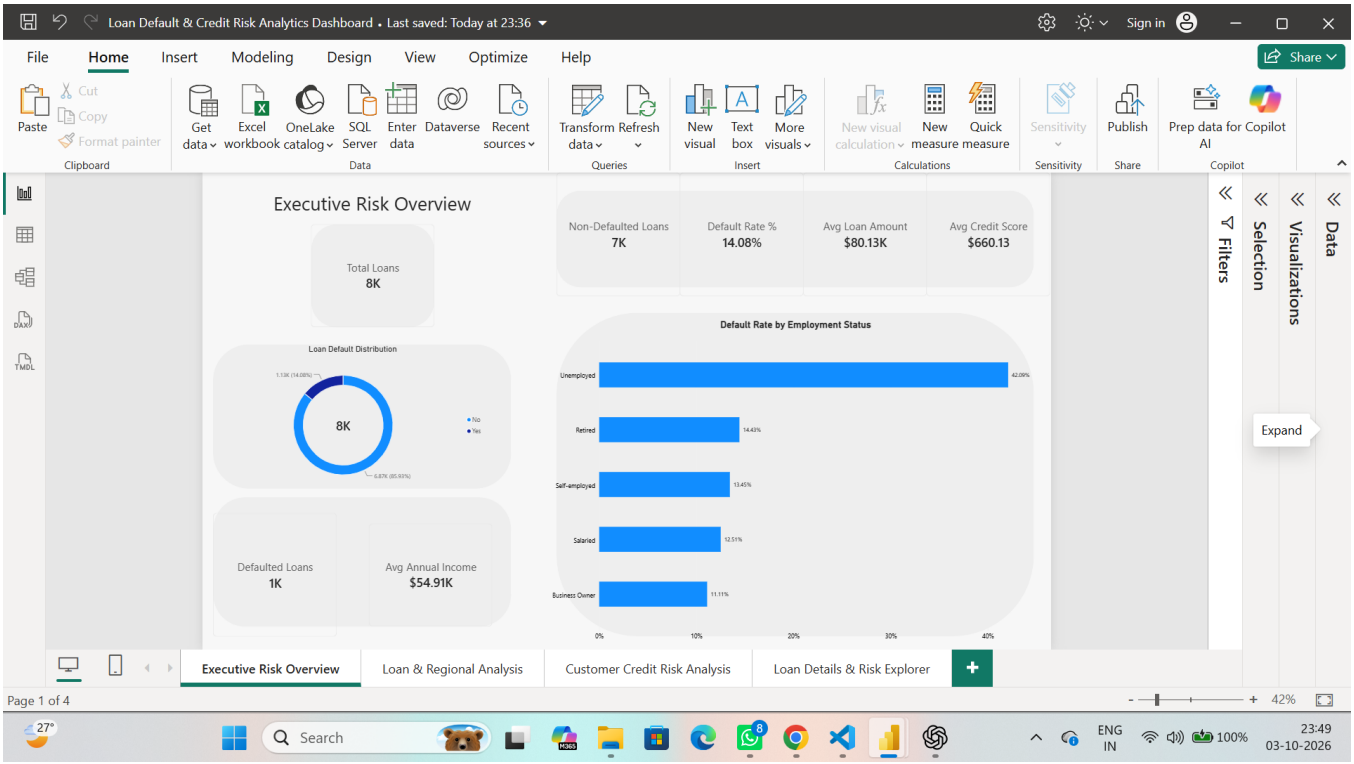
Income Group Default Rates

Income Group	Default Rate
Low Income	24.91%
Medium Income	13.51%
High Income	9.59%
Very High Income	5.77%

These are descriptive results from the supplied dataset and do not establish causal relationships.

4. Power BI — Executive Risk Overview

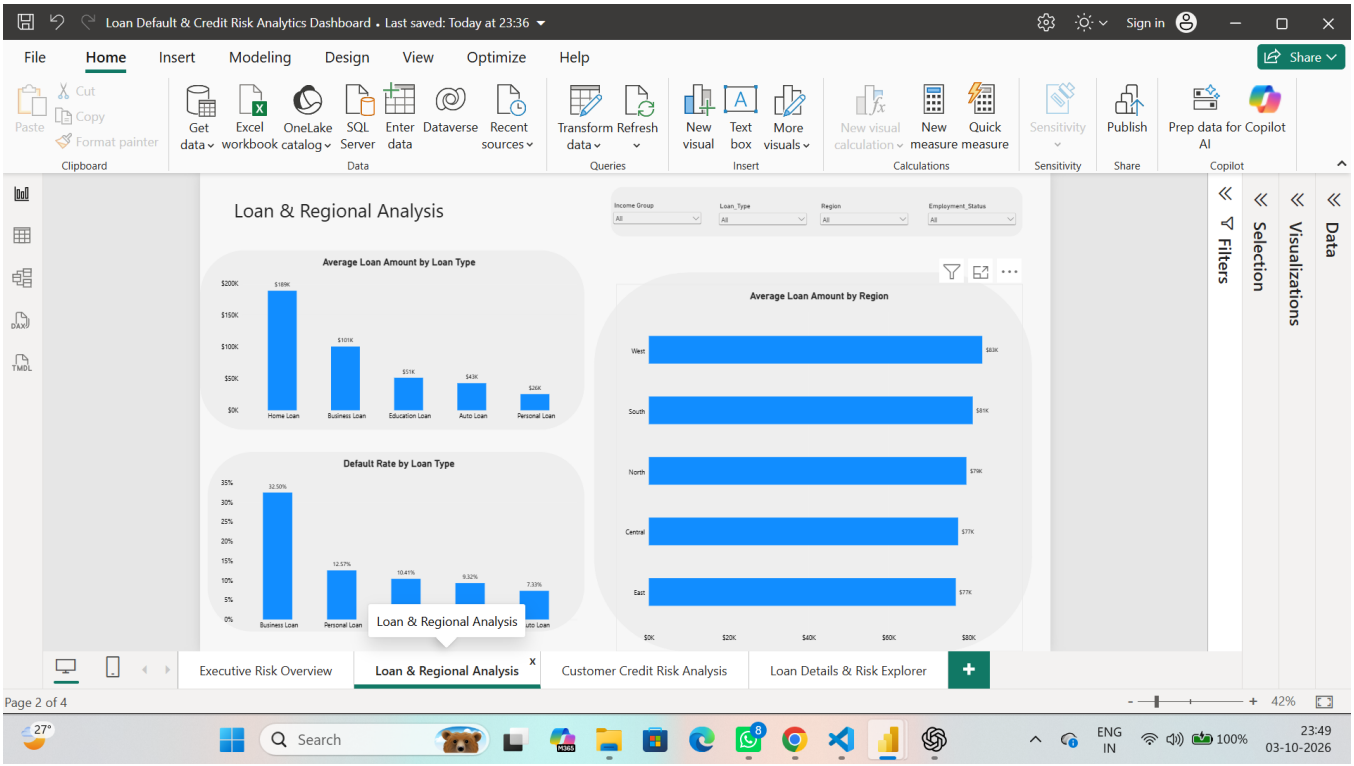
KPI cards, loan default distribution, and employment-status default rates.



Supplied Power BI dashboard screenshot.

5. Power BI — Loan & Regional Analysis

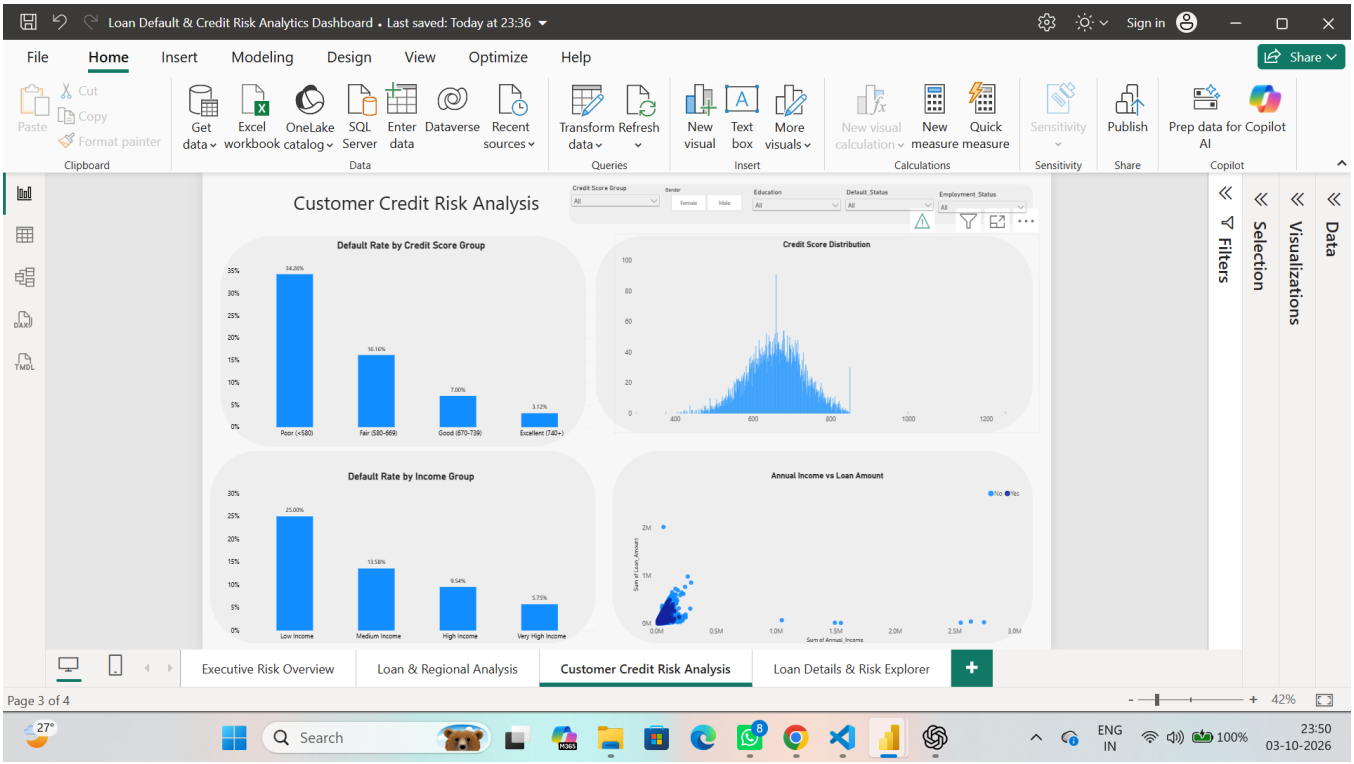
Average loan amount by loan type and region, plus loan-type default rates.



Supplied Power BI dashboard screenshot.

6. Power BI — Customer Credit Risk Analysis

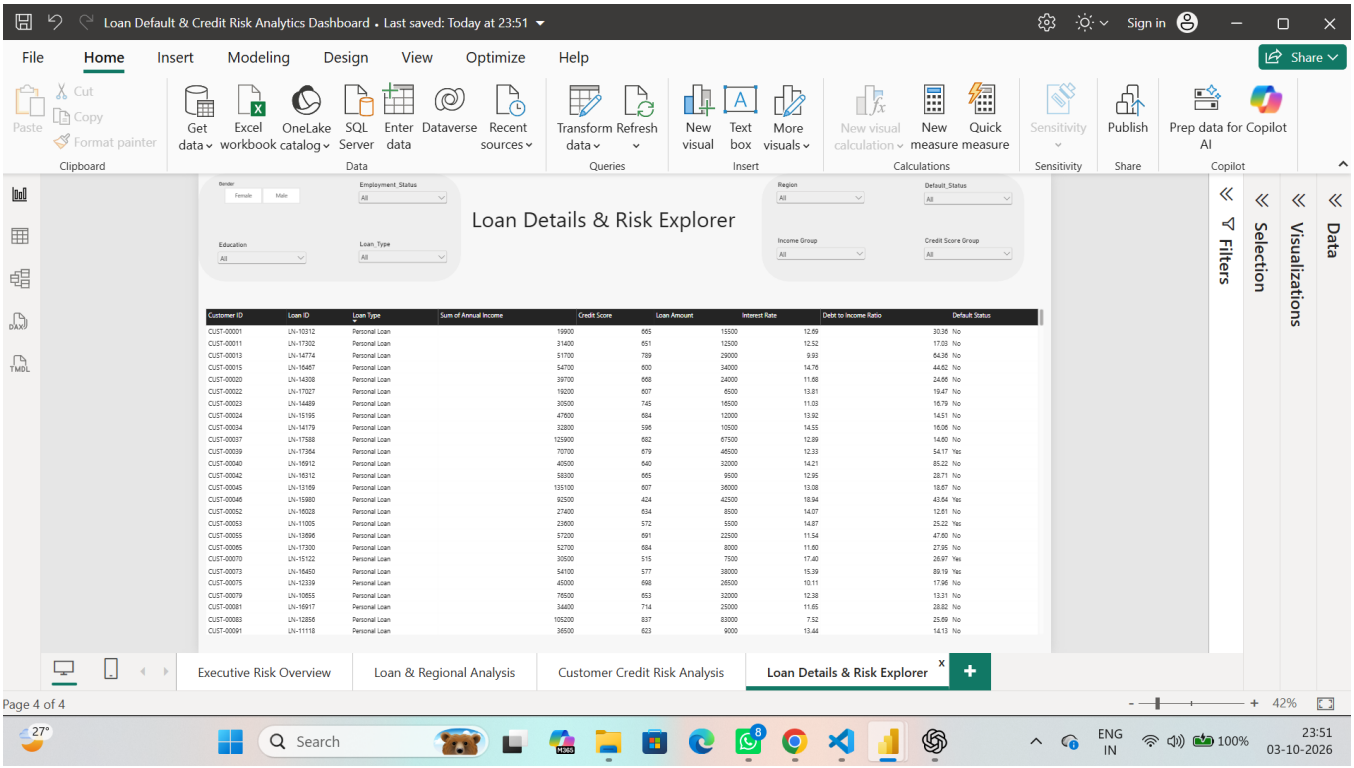
Credit score distribution, credit-score groups, income groups, and income-versus-loan analysis.



Supplied Power BI dashboard screenshot.

7. Power BI — Loan Details & Risk Explorer

Detailed loan-level table with interactive slicers.



Supplied Power BI dashboard screenshot.

8. Conclusion & Deliverables

The project demonstrates a complete analytics workflow from cleaned loan data to a four-page interactive Power BI dashboard. The dashboard provides executive-level metrics, loan and regional analysis, customer credit-risk analysis, and detailed loan exploration.

Deliverables

- Loan_Credit_Risk_Cleaned.csv
- Loan_Credit_Risk_Analysis.py
- Loan_Default_Credit_Risk_Analysis_Report.pdf
- README_Loan_Default_Credit_Risk.md
- Four-page Power BI dashboard

Tools: Python • Pandas • Matplotlib • Seaborn • Jupyter/VS Code • Power BI • DAX

For educational and analytical use. Dataset relationships are descriptive and should not be treated as causal or as an automated lending decision.